

DBMS Lecture 8: Super Key Mechanics

Comprehensive Guide to Candidate Keys, Super Keys & GATE Counting Formulas

1 1. Introduction & Purpose of Keys

In Relational Database Management Systems (RDBMS), a table represents a mathematical relation—a collection of tuples (rows). To ensure data integrity, distinguish tuples, and form relational connections between tables, DBMS relies on **Keys**.

What is a Key? ★

A **Key** is an attribute or a minimal set of attributes used to uniquely identify a specific tuple (row) within a relation (table).

Motivational Example: Resolving Record Ambiguity

Consider a student database table lacking a designated unique identifier key:

Roll_No	Name	Age
1	Reddy	20
2	Reddy	20
3	Amit	21

Notice that using only {Name, Age} is insufficient because two different students share the identical combination ('Reddy', 20). Only the Roll_No column strictly guarantees unique identification of each tuple.

2 2. Candidate Key

What is a Candidate Key?

A **Candidate Key** is a **minimal set of attributes** capable of uniquely identifying every tuple in a relation.

A candidate key must satisfy two fundamental mathematical properties:

1. **Uniqueness Property:** No two tuples in the relation can have identical values for the candidate key attribute set.
2. **Minimality Property:** No proper subset of the candidate key can uniquely identify the tuples. Removing even a single attribute breaks the uniqueness constraint.

Example

In Student(Roll_No, Name, Age), if Roll_No is guaranteed unique:

Candidate Key = {Roll_No}

It is **minimal** because removing Roll_No leaves an empty set $\emptyset$, which cannot identify tuples.

3 3. Super Key ★★

What is a Super Key?

A **Super Key** is **any set of attributes** that uniquely identifies every tuple in a relation.

The simplest intuitive way to define a super key is:

Super Key = Candidate Key + Zero or More Additional Attributes

Example Walkthrough

Given Student(Roll_No, Name, Age) where Roll_No is the Candidate Key:

- {Roll_No} $\implies$ Super Key (Zero extra attributes added)
- {Roll_No, Name} $\implies$ Super Key (Contains Roll_No)
- {Roll_No, Age} $\implies$ Super Key (Contains Roll_No)
- {Roll_No, Name, Age} $\implies$ Super Key (Contains Roll_No)

Every superset of a candidate key is automatically a valid super key because it retains the foundational unique identifier attribute (Roll_No).

4 4. Candidate Key vs. Super Key Matrix

Table 1: In-Depth Structural Comparison Matrix

Attribute / Property	Candidate Key	Super Key
Minimality	Strictly Minimal (No redundant attributes)	Not Necessarily Minimal (May contain extra attributes)
Uniqueness	Guarantees unique tuple identification	Guarantees unique tuple identification
Redundant Fields	Cannot contain unnecessary attributes	Can contain unnecessary extra attributes
Relationship Rule	Every Candidate Key is a Super Key	Not every Super Key is a Candidate Key
Role in Design	Baseline candidate for Primary Key selection	Functional dependency target for Boyce-Codd Normal Form (BCNF)

5 5. The Golden Rule & Set Inclusion Topology ★

The Golden Principle

Every Candidate Key is a Super Key, but every Super Key is NOT a Candidate Key.

Set Topological Hierarchy

Super Keys Set (SK) ⊃ Candidate Keys Set (CK) ⊃ Primary Key (PK)

Candidate Keys represent the **irreducible kernel** (minimal subset) within the broader universe of Super Keys.

6. Why Uniqueness Fails for Non-Keys

Suppose we test the attribute combination {Name, Age} on the following relation:

Roll_No	Name	Age
1	Reddy	20
2	Reddy	20

- Evaluating {Name, Age} yields ('Reddy', 20) for both row 1 and row 2.
- **Conclusion:** {Name, Age} **is NOT a Super Key** because it cannot resolve tuple collision.
- Evaluating {Roll_No, Name} yields (1, 'Reddy') vs (2, 'Reddy').
- **Conclusion:** {Roll_No, Name} **is a valid Super Key**.

7. Comprehensive Attribute Subset Evaluation Table

Given relation Student(Roll_No, Name, Age) with Candidate Key = {Roll_No}:

Table 2: Complete Classification of All Possible Subsets ($2^3 = 8$ Subsets)

Attribute Set	Super Key?	Candidate Key?	Reason / Technical Justification
{Roll_No}	YES	YES	Unique and minimal.
{Roll_No, Name}	YES	NO	Super key, but contains redundant attribute Name.
{Roll_No, Age}	YES	NO	Super key, but contains redundant attribute Age.
{Roll_No, Name, Age}	YES	NO	Super key, but contains 2 redundant attributes.
{Name}	NO	NO	Names can duplicate.
{Age}	NO	NO	Ages can duplicate.
{Name, Age}	NO	NO	Combined name and age can duplicate.
∅ (Empty Set)	NO	NO	Cannot identify tuples.

8. Numerical Mechanics: Counting Super Keys ★★

A classic category of questions in **GATE, UGC NET, and PSU Computer Science exams** involves calculating the exact number of valid super keys given a relation schema and a set of candidate keys.

Case 1: Single Single-Attribute Candidate Key

Let relation R have N attributes: $R(A_1, A_2, A_3, \dots, A_n)$. Assume A_1 is the sole Candidate Key.

Formula: Single Single-Attribute Candidate Key

$$\text{Total Super Keys} = 2^{N-1}$$

Mathematical Derivation:

1. Every super key **must** contain A_1 (1 choice: compulsory).
2. For each of the remaining $(N - 1)$ attributes, we have 2 independent choices: include it or exclude it.
3. By product rule of combinatorics: $1 \times \underbrace{2 \times 2 \times \dots \times 2}_{(N-1) \text{ times}} = 2^{N-1}$.

Example 8.1

Let $R(A, B, C, D)$ ($N = 4$) with Candidate Key = $\{A\}$.

$$\text{Total Super Keys} = 2^{4-1} = 2^3 = 8$$

The 8 super keys are explicitly:

$$\{A\}, \{A, B\}, \{A, C\}, \{A, D\}, \{A, B, C\}, \{A, B, D\}, \{A, C, D\}, \{A, B, C, D\}$$

Case 2: Two Single-Attribute Candidate Keys

Let relation $R(A_1, A_2, A_3, \dots, A_n)$ have two disjoint single-attribute candidate keys: $K_1 = \{A_1\}$ and $K_2 = \{A_2\}$.

Formula: Two Single-Attribute Candidate Keys

$$\text{Total Super Keys} = 2^{N-1} + 2^{N-1} - 2^{N-2} = 3 \times 2^{N-2}$$

Derivation via Principle of Inclusion-Exclusion (PIE):

- Super keys containing A_1 : $|S_1| = 2^{N-1}$
- Super keys containing A_2 : $|S_2| = 2^{N-1}$
- Intersection (Super keys containing **both** A_1 and A_2): $|S_1 \cap S_2| = 2^{N-2}$

$$|S_1 \cup S_2| = |S_1| + |S_2| - |S_1 \cap S_2| = 2^{N-1} + 2^{N-1} - 2^{N-2} = 3 \cdot 2^{N-2}$$

Example 8.2

Let $R(A, B, C, D)$ ($N = 4$) with Candidate Keys $K_1 = \{A\}$ and $K_2 = \{B\}$.

$$\text{Total Super Keys} = 2^{4-1} + 2^{4-1} - 2^{4-2} = 8 + 8 - 4 = 12$$

Case 3: Single Composite Candidate Key

Let relation $R(A_1, A_2, \dots, A_n)$ have one composite candidate key of size K (containing K attributes).

Formula: Single Composite Candidate Key of Size K

$$\text{Total Super Keys} = 2^{N-K}$$

Example 8.3

Let $R(A, B, C, D, E)$ ($N = 5$) with Candidate Key = $\{A, B\}$ (size $K = 2$).

$$\text{Total Super Keys} = 2^{5-2} = 2^3 = 8$$

The super keys are: $\{A, B\}, \{A, B, C\}, \{A, B, D\}, \{A, B, E\}, \{A, B, C, D\}, \{A, B, C, E\}, \{A, B, D, E\}, \{A, B, C, D, E\}$

Case 4: Two Disjoint Composite Candidate Keys

Let $R(A_1, A_2, \dots, A_n)$ have candidate keys K_1 (size k_1) and K_2 (size k_2) such that $K_1 \cap K_2 = \emptyset$.

Formula: Two Disjoint Composite Candidate Keys

$$\text{Total Super Keys} = 2^{N-k_1} + 2^{N-k_2} - 2^{N-(k_1+k_2)}$$

9. General Inclusion-Exclusion Formula ★ ★ ★

When dealing with arbitrary, overlapping, or multi-attribute candidate keys $K_1, K_2, \dots, K_m$, apply the generalized ****Inclusion-Exclusion Principle****:

General Multi-Key Formula (Two Keys K_1, K_2)

$$|SK| = 2^{N-|K_1|} + 2^{N-|K_2|} - 2^{N-|K_1 \cup K_2|}$$

Where:

- N = Total number of attributes in relation R
- $|K_1|$ = Number of attributes in candidate key 1
- $|K_2|$ = Number of attributes in candidate key 2
- $|K_1 \cup K_2|$ = Total number of distinct attributes present in $K_1 \cup K_2$

Example 9.1: Overlapping Candidate Keys

Let $R(A, B, C, D)$ ($N = 4$) with candidate keys $K_1 = \{A, B\}$ and $K_2 = \{B, C\}$.

1. $|K_1| = 2 \implies 2^{4-2} = 4$
2. $|K_2| = 2 \implies 2^{4-2} = 4$
3. $K_1 \cup K_2 = \{A, B, C\} \implies |K_1 \cup K_2| = 3 \implies 2^{4-3} = 2$

$$\text{Total Super Keys} = 4 + 4 - 2 = 6$$

10 10. Common Pitfalls & Misconceptions

Exam Caution: Critical Misconceptions

- **Pitfall 1:** Thinking any attribute combination is a super key.
→ **Correction:** A combination is a super key **only** if it contains at least one full candidate key.
- **Pitfall 2:** Forgetting to subtract the intersection when counting super keys for multiple candidate keys.
→ **Correction:** Sets containing multiple candidate keys will be double-counted if inclusion-exclusion is ignored.
- **Pitfall 3:** Assuming candidate keys cannot overlap.
→ **Correction:** Candidate keys frequently share attributes (e.g., $\{A, B\}$ and $\{B, C\}$). Use $|K_1 \cup K_2|$ in the exponent.

11 11. Master Comparison Matrix: Super vs. Candidate vs. Primary Key

Table 3: Three-Way Comparison of Key Constructs

Property	Super Key	Candidate Key	Primary Key
Uniqueness	Required	Required	Required
Minimality	Not required	Strictly Required	Strictly Required
Allowed Quantity	Many (2^{N-K})	One or more (≥ 1)	Exactly One per table
Extra Fields	Allowed	Strictly forbidden	Strictly forbidden
NULL Permitted	Context-dependent	Generally forbidden	Strictly Forbidden (NOT NULL)
Selection	Foundational set	Minimal set filtered from SK	Single choice selected from CKs

12 12. Target GATE & Interview Q&A

Q1. Define a Super Key in one sentence.

A super key is any set of attributes within a relation that can uniquely identify every tuple.

Q2. How does a Candidate Key differ from a Super Key?

A candidate key is a minimal super key; it contains no redundant attributes that can be removed while maintaining uniqueness.

Q3. Given $R(A, B, C, D, E)$ with Candidate Key $\{A\}$, how many super keys exist?

Here $N = 5, K = 1$. Number of super keys = $2^{5-1} = 2^4 = 16$.

Q4. Given $R(A, B, C, D)$ with Candidate Keys $\{A\}$ and $\{B, C\}$, find total super keys.

$N = 4$.

- For $K_1 = \{A\}$ ($k_1 = 1$): $2^{4-1} = 8$
- For $K_2 = \{B, C\}$ ($k_2 = 2$): $2^{4-2} = 4$
- Intersection $K_1 \cup K_2 = \{A, B, C\}$ ($|K_1 \cup K_2| = 3$): $2^{4-3} = 2$

Total = $8 + 4 - 2 = 10$.

Q5. Is the empty set $\emptyset$ ever a valid Super Key?

No, an empty set cannot hold tuple values to establish uniqueness.

Q6. What is the maximum possible number of super keys for a relation with N attributes?

If a single attribute is a candidate key, maximum super keys = 2^{N-1} . If all attributes individually are candidate keys, maximum super keys = $2^N - 1$.

13 13. One-Minute Revision & Exam Cheat Sheet

Ultimate Super Key Revision Guide

SUPER KEY $\Rightarrow$ Any attribute set providing unique tuple identification

CANDIDATE KEY $\Rightarrow$ Minimal Super Key (No extra fields)

PRIMARY KEY $\Rightarrow$ The single Candidate Key chosen by DB designer

Quick Counting Formulas Matrix for Exams: • 1 CK of size 1 $\Rightarrow 2^{N-1}$

• 1 CK of size K $\Rightarrow 2^{N-K}$ • 2 Disjoint CKs of size 1 $\Rightarrow 3 \cdot 2^{N-2}$

• PIE General Rule $\Rightarrow |SK| = 2^{N-|K_1|} + 2^{N-|K_2|} - 2^{N-|K_1 \cup K_2|}$

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